

Lab 2.1 – Image Acquisition

At the end of this lab exercise, Participants will be able to:

- Acquire a good image on the camera

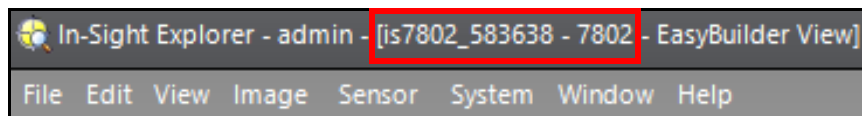
The Participant will utilize the following In-Sight Functions to successfully complete this exercise:

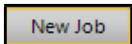
- Easy Builder User Interface Application Step 1
 - Set up Image

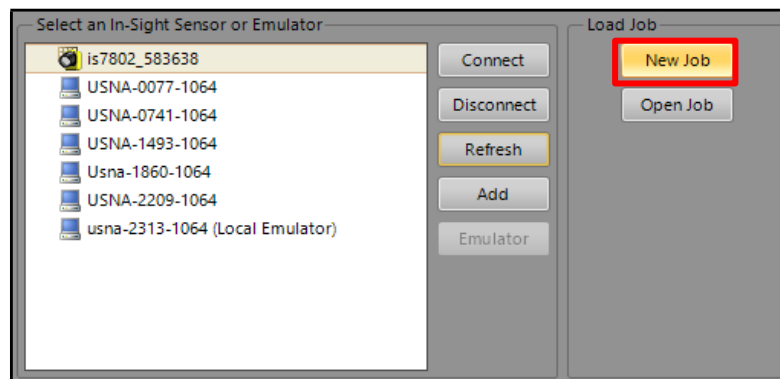
Follow the steps below to complete the lab exercise:

1. Create a folder on your desktop named **InSightClass** with the date appended e.g. **InSightClass012720**. This is where you will save the jobs that you develop in the lab exercises.
2. Click the **Get Connected**  button. Select the camera and click the **Connect** button.

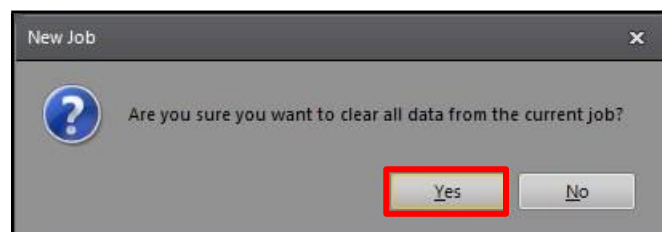
NOTE: The top border of ISE will indicate what camera you are logged onto. Confirm that you are logged onto your camera.

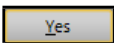


3. Click the **New Job**  button in the **Load Job** section.

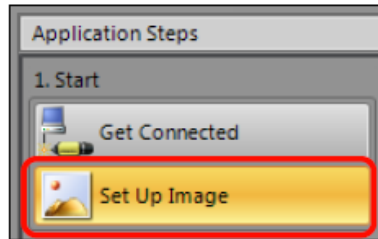


The **'Are you sure you want to clear all data from the current job?'** disclaimer displays.



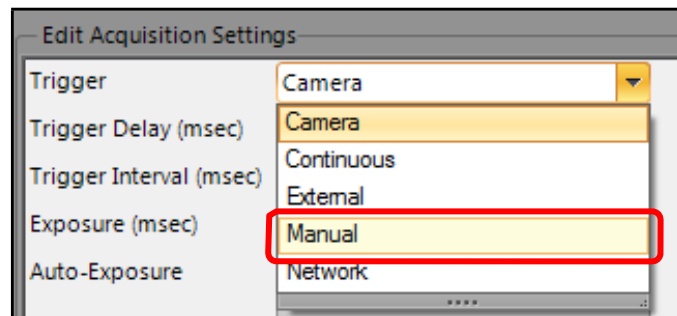
4. Click the **Yes**  button.

5. Place the plate representing a good part under the camera. Click the **Set Up Image**  button under Application Steps.



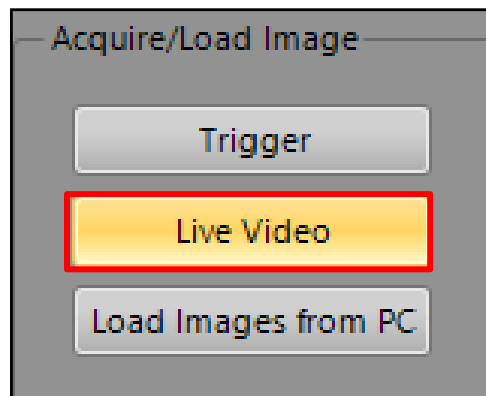
Follow the steps below if acquiring images with an **In-Sight Camera**:

6. Ensure the following **Edit Acquisition Settings**, where applicable:
- Trigger: Manual (see below)
 - Built-in ring light: Off (disabled) to reduce glare
 - Exposure: Disable automatic, set for manual
 - Focus Region: Position region on plate and click OK.



NOTE: On a three-dimensional part, you would need to decide where to focus.

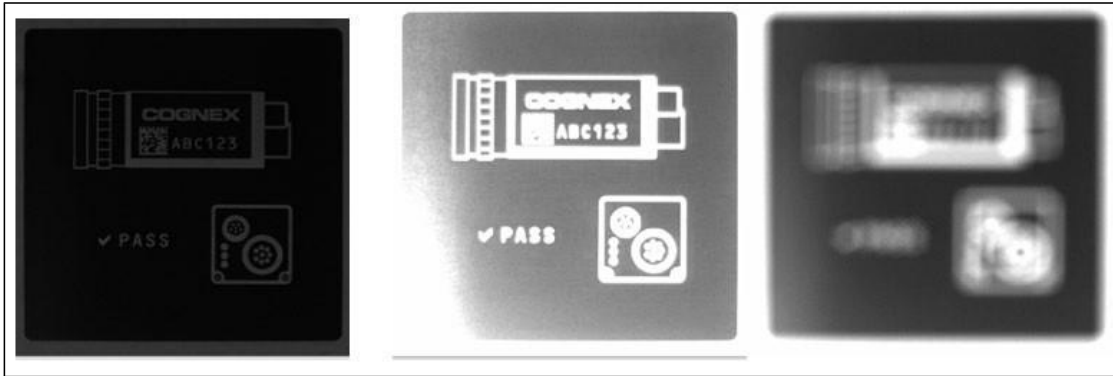
7. Click the **Live Video**  button in the **Acquire/Load Image** box to start a live image.



8. Move the part under the camera to confirm the image is updating. (If light levels are too low, you might not see the image yet.)
9. Next, you will adjust settings to obtain the best image possible. This means sharp focus and good contrast. The procedure depends on whether you have a manually adjustable lens or an autofocus lens (available on 7xxx series).

Here are examples of poor quality images:

a. *Too Dark*, b. *Too Light*, and c. *Too Blurry*.

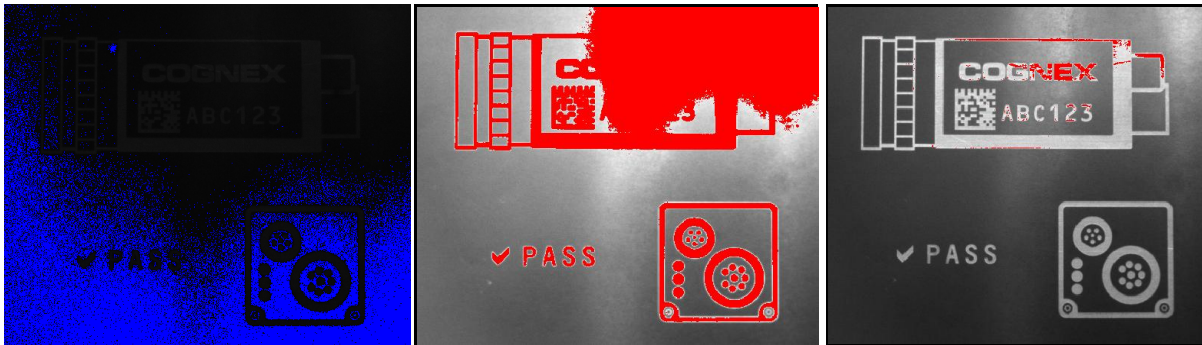


Here is an example of a good quality image:

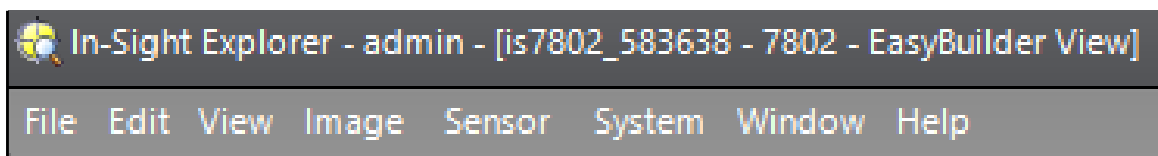


10. First, let's get good contrast:
 - a. 5000, 7000 and Micro cameras with C-mount lens
Adjust the aperture ring to obtain a good range of light and dark (contrast)
 - b. 7000 cameras with Autofocus lens
In Edit Acquisition Settings, adjust the Exposure value manually to obtain good contrast.
 11. Next, let's get good focus.
 - a. 5000, 7000 and Micro cameras with C-mount lens
Adjust the focus ring to obtain a sharp focus.
 - b. 7000 cameras with Autofocus lens. Go into Live Video and Adjust the focus by using the Autofocus  button.
- NOTE:** *The button is in the lower right hand corner of the live video view.*
12. Review the image capture, looking for:
 - a. Maximum difference between light and dark areas (good contrast).
 - b. Sharp features (not blurry)
 13. To verify the writing on the plate is dark and the metallic background light on your image, click the **Show Brightness Feedback**  button (top icon bar) to assist with this.

NOTE: Too much blue may mean that the image is too dark and too much red may mean that the image is too light. This is meant as a guide; blue or red does not necessarily mean a bad image.



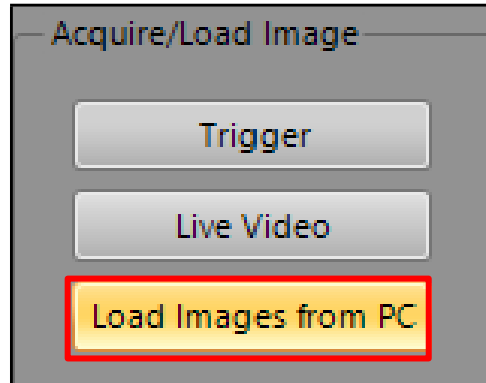
14. Once you have a good image, click **Live Video** to exit Live mode.
15. If the cameras in the training room are on a network, find another In-Sight system in the room and ask its users if it is OK for you to try and log into it.
16. With the Set Up Image button selected, activate **Live Mode** on their system to verify that you have logged into the correct system. Offer them the same courtesy.
17. Browse through the various drop-down menus in In-Sight Explorer to compare what is available on both the Icon bar and within the Menu bar.



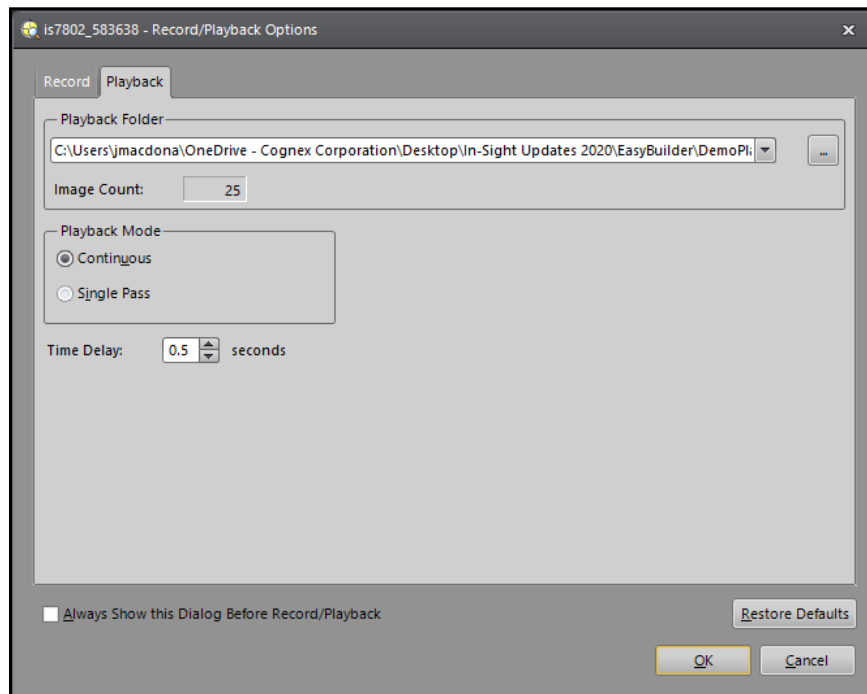
18. Save the job as **MyFocus** in the folder that you created in Step 1.

Follow the steps below if using an **Image Database**:

6. Click the **Load Images from PC** button in the **Acquire/Load Image** box to start a live image.



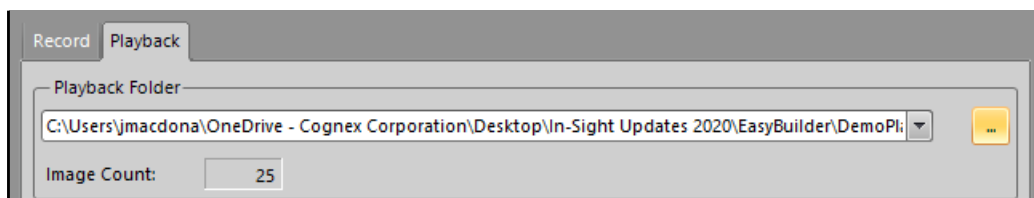
The **Record/Playback Options** dialog displays.



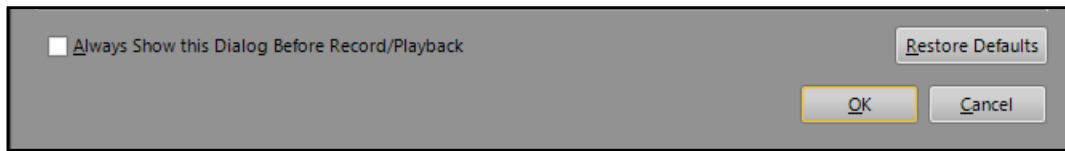
The **Record/Playback Options** dialog displays.

7. From the **Playback** tab click the elipsis to browse for the images.

NOTE: *The instructor will tell you where to find the images to complete the lab exercises.*



8. Click the **OK** button.



9. Save the job as **MyFocus** in the folder that you created in Step 1.

Lab 2.2 – Scale Calibration

At the end of this lab exercise, Participants will be able to:

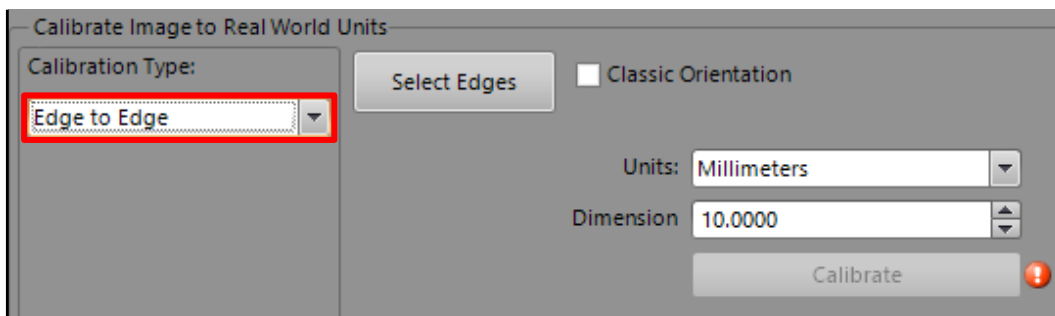
- Set up scale calibration and grid calibration to obtain real world values
- Convert pixel values to real world values by using a feature of known length.

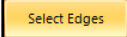
The Participant will utilize the following In-Sight Functions to successfully complete this exercise:

- Easy Builder User Interface Application Step 1
 - Set up Image
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Follow the steps below to complete the lab exercise:

1. Continue with the **MyFocus** from the previous lab exercise.
NOTE: *Confirm that your demo plate is on the PASS side and that you are not in Live mode.*
2. Save the job as **MyScaleCalibration** in the folder on the desktop created in Lab #2.
3. Select **Edge to Edge** as the Calibration Type in the **Calibrate Image to Real World Units** box.

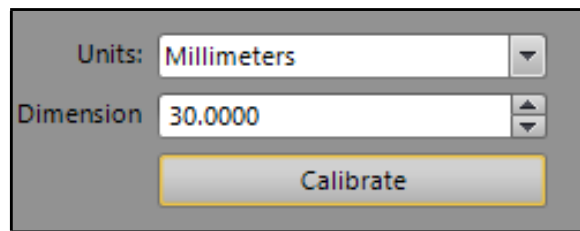


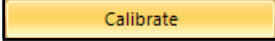
4. Click the **Select Edges**  button.
5. Click the top of the Smart Feature, drag your mouse and click at the bottom of the Smart Feature graphic to use the distance as your scale reference.



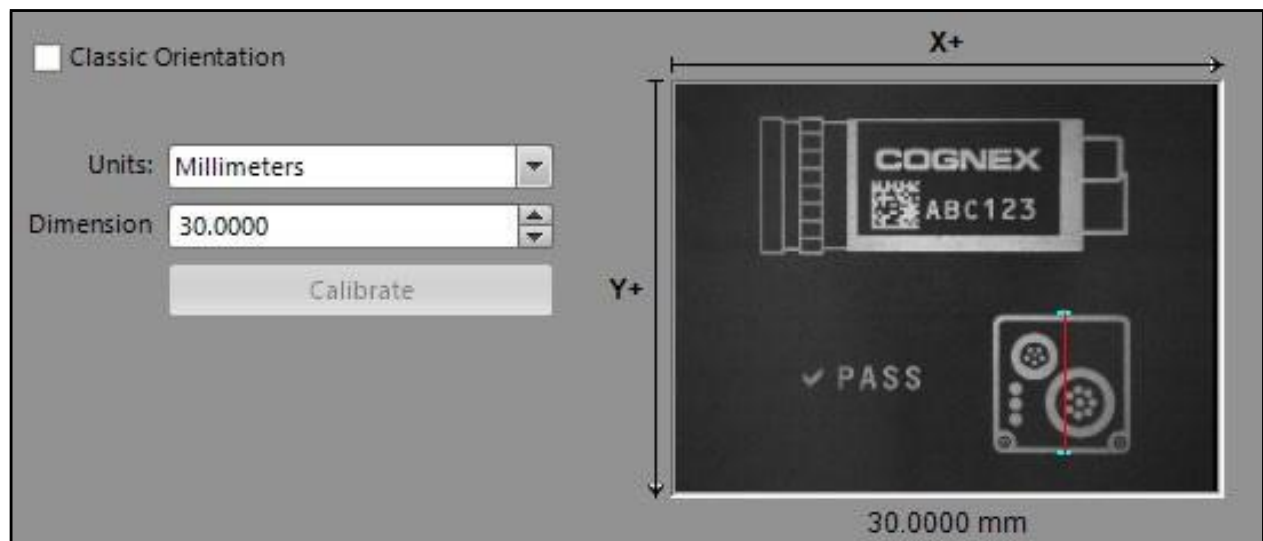
6. Set **Units** to Millimeters and **Dimension** to **30 (Might be DIFFERENT from yours)**.

NOTE: This denotes that there are 30mm between the top and bottom edges selected.

A small dialog box with a grey background. It contains a 'Units' dropdown menu set to 'Millimeters', a 'Dimension' input field with the value '30.0000', and a 'Calibrate' button at the bottom.

7. Click the **Calibrate**  button.

NOTE: From this point forward all measurements will be reported in millimeters.



8. Save the job.

Lab 2.3 – Grid Calibration

At the end of this lab exercise, Participants will be able to:

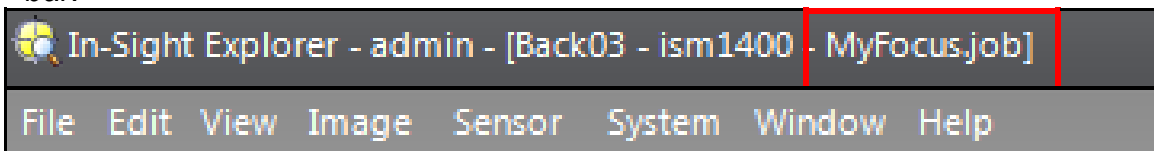
- Convert pixel values to real world values by using a preprinted grid and calibration wizard.

The Participant will utilize the following In-Sight Functions to successfully complete this exercise:

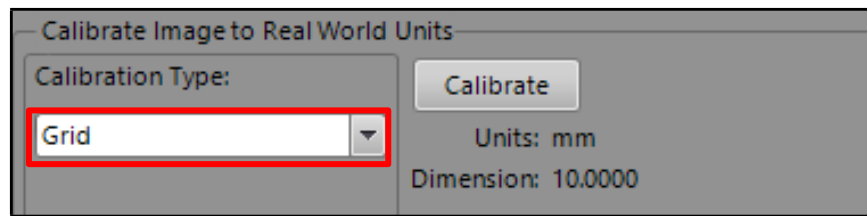
- Easy Builder User Interface Application Step 1
 - Set up Image

Follow the steps below to complete the lab exercise:

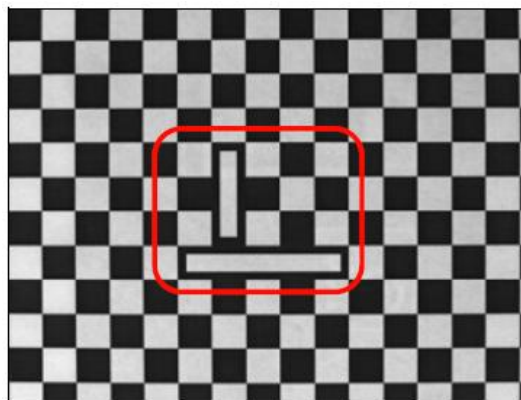
1. Load the **MyFocus** job from the first lab exercise.
2. Save the job as **MyGridCalibration** in the folder on the desktop created in Lab #2.
NOTE: Confirm that you have loaded the correct job by looking at the window's title bar.

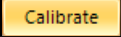


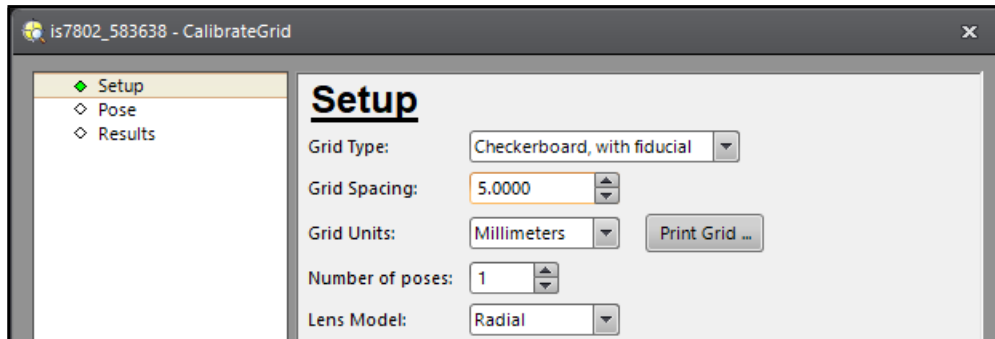
3. Select **Grid** as the Calibration Type in the **Calibrate Image to Real World Units** box.



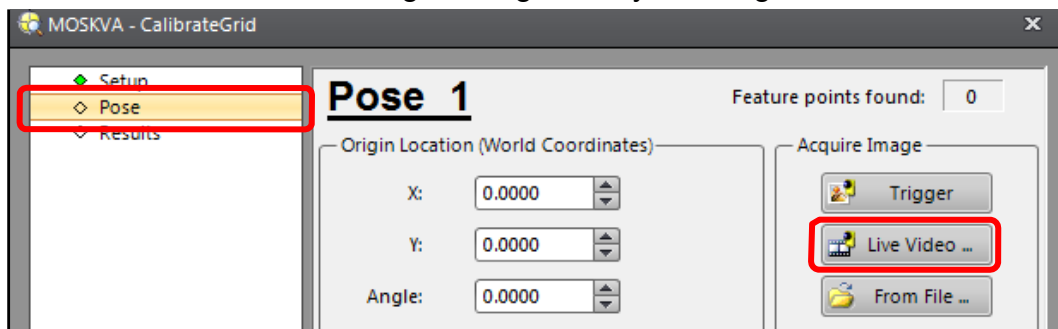
NOTE: Use the checkerboard grid with fiducial (XY locator bars). Your instructor will tell you which grid to use and the value to use for **Grid Spacing**. If it is the grid in the image booklet it is 5 mm.



4. Click the **Calibrate**  button.
The **CalibrateGrid** dialog box opens.



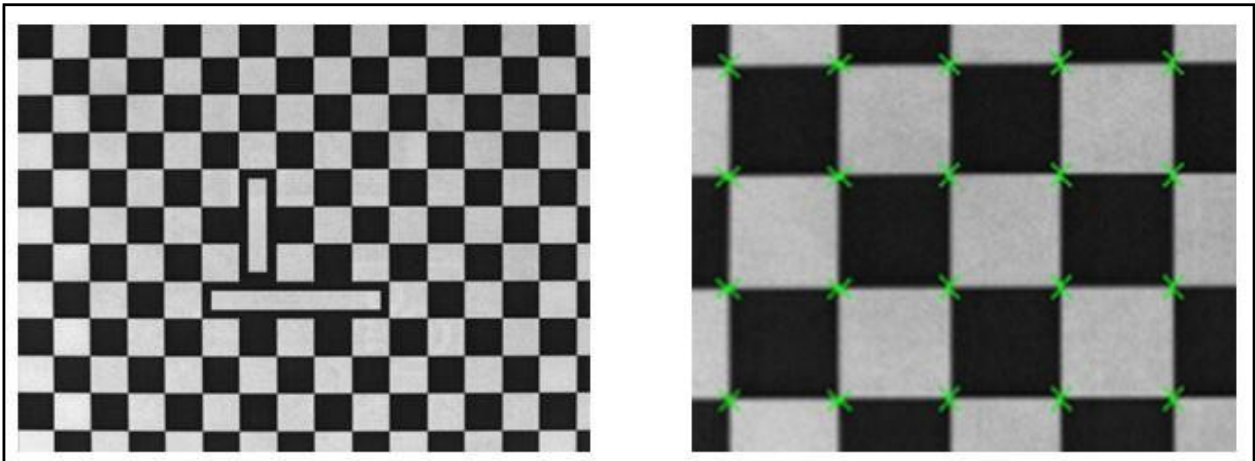
5. Select the settings that match the checkerboard image calibration plate as shown above.
 - **Grid Type:** *Checkerboard, with fiducial*
 - **Grid Spacing:** *5.0*
 - **Grid Units:** *Millimeters*
 - **Number of poses:** *1*
 - **Lens Model:** *Radial*
6. Click the **Pose** step
If using an In-Sight Camera:
 - Click the **Live Video**  button.If using an Image Set:
 - Select the calibration grid image from your image set.



7. Place the checkerboard grid under the camera and adjust for contrast and sharpness if necessary.
8. Click anywhere in the image to stop Live mode and check for the green **X** marks in the checkerboard image.

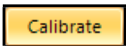
NOTE: If there are no green marks go back to step 5.

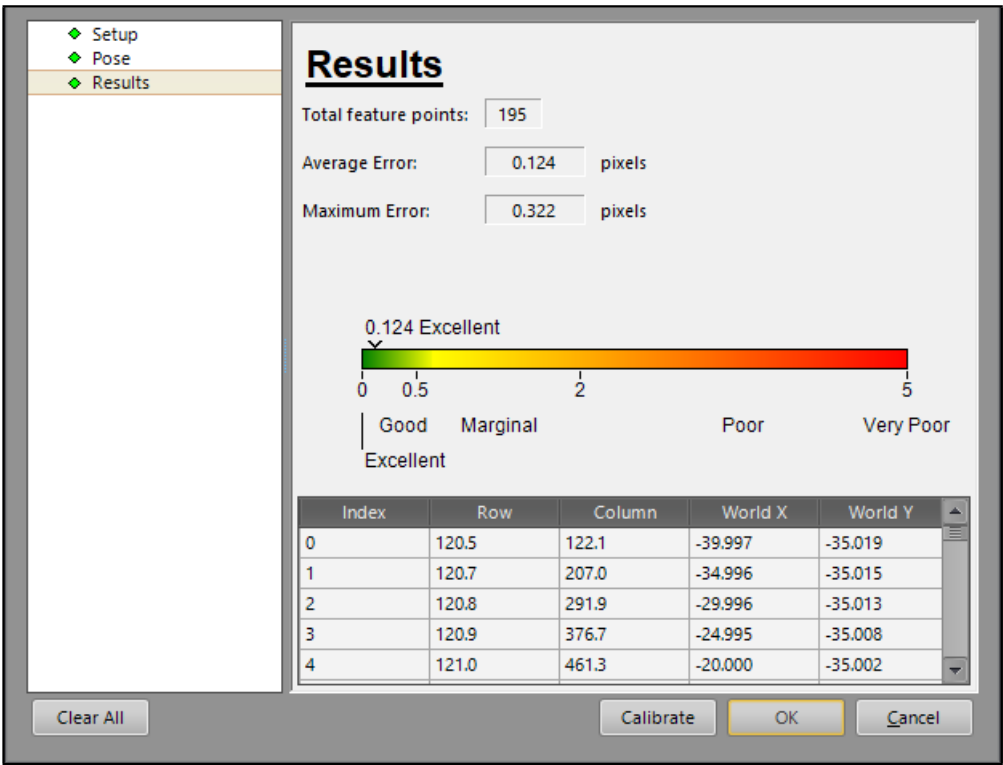
- a. Live Image
- b. Static Image (Zoomed in)



9. Confirm the **Features points found** is *not* 0.



10. Click the **Calibrate**  button.
The **Calibration** results display.



11. Click the **OK**  button to complete the calibration procedure.
12. Save the job.